

User Flow & Sequence Document - HMS Patient Journey

Version 1.0 | Core Workflow Definition

Prepared by Pratyaksh Lutare as the master user flow reference for backend and frontend developers.

DOCUMENT PURPOSE

This document translates the visual Patient Journey flowchart into a structured, step-by-step technical user flow. Since visual diagrams can be difficult to translate directly into code, this document breaks down the flow into logical nodes, decision trees, and system actions. Developers should use this to build state machines, frontend routing, and API sequences.

1. High-Level Patient Journey Flow

The core flow follows a linear progression with conditional branches based on clinical decisions (OPD vs. IPD) and system-triggered background actions.

- 1. **Entry & Identity** (Registration / Search)
- 2. **Visit Initiation** (Appointment / Walk-in → Check-in)
- 3. **Queue & Triage** (Waiting Room → Vitals)
- 4. **Consultation** (Doctor Examination → EMR → Orders)
- 5. **Post-Consultation Routing** (Pharmacy / Labs / Admission)
- 6. **Billing & Settlement** (Invoicing → Payment)
- 7. **Exit & Follow-up** (Discharge → Notifications)

2. Detailed Developer User Flows (Node-by-Node)

Phase 1: Entry & Identity Management

Step	Actor	Action / UI Node	System Logic / Backend API
1.1	Receptionist / Patient	Initiate Visit	Check if patient is New or Existing.
1.2a	Receptionist	[Branch: New	Create Patient

Step	Actor	Action / UI Node	System Logic / Backend API
		Patient] Fill Registration Form	Record API. Generate unique UHID .
1.2b	Receptionist	[Branch: Existing] Search Patient	Search API (by Phone, Name, or UHID). Fetch Record.
1.3	System	Trigger Welcome Message	[System Action] Fire SMS/Email via Notification Service.

Phase 2: Appointment & Visit Initiation

Step	Actor	Action / UI Node	System Logic / Backend API
2.1	Receptionist / Patient	Select Doctor & Department	Fetch Doctor Availability API.
2.2	Receptionist	Book Slot or Walk-in	Create Appointment API. Set status to SCHEDULED or WALK_IN.
2.3	Receptionist	Mark as Arrived (Check-in)	Update Appointment Status to CHECKED_IN. Generate Queue Token.
2.4	System	Update Queue	[System Action] Push token to Doctor's Queue Dashboard via WebSocket/Polling.

Phase 3: Clinical Consultation (The Doctor Flow)

Step	Actor	Action / UI Node	System Logic / Backend API
3.1	Nurse (Optional)	Pre-Consultation (Vitals)	Log Vitals API (BP, Weight, Temp). Attach to current Visit ID.
3.2	Doctor	Select Patient from Queue	Update Appointment Status to IN_CONSULTATION. Load EMR history.
3.3	Doctor	Add Clinical Notes & Diagnosis	Update EMR API. Save Chief Complaints and ICD/Custom Diagnosis codes.
3.4	Doctor	Create Treatment Plan	Decision Node: Order Labs? Prescribe Meds? Recommend Admission?
3.5	Doctor	Submit Consultation	Update Appointment Status to CONSULTATION_DONE. Trigger downstream modules.

Phase 4: Post-Consultation Branching (Conditional)

Depending on step 3.4, the flow splits into parallel branches.

Branch	Actor	Action Flow	System Logic / Backend API
Lab Orders	Lab Tech	View Order → Collect Sample → Process → Upload Report	Create Lab Order API. Update Status to REPORT_READY. [System Action] Notify Doctor & Patient.

Branch	Actor	Action Flow	System Logic / Backend API
Pharmacy	Pharmacist	View e-Prescription → Check Inventory → Dispense	Fetch Prescription API. Deduct stock from Inventory API. Send charge to Billing.
Admission (IPD)	IPD Manager	View Request → Allocate Ward/Bed → Admit Patient	Convert OPD to IPD API. Update Bed Status API to OCCUPIED.

Phase 5: Billing & Settlement

Step	Actor	Action / UI Node	System Logic / Backend API
5.1	Billing Staff	Load Patient Ledger	Fetch all unbilled charges (Consultation, Labs, Pharmacy, Room).
5.2	Billing Staff	Apply Insurance/Discount (Optional)	Calculate final payable amount API. Update claim status if TPA involved.
5.3	Billing Staff / Patient	Process Payment	Initiate Payment Gateway API. Record transaction success/failure.
5.4	System	Generate Invoice & Receipt	[System Action] Generate PDF. Email to patient. Clear outstanding balance.

3. System Automations & Background Processes

Throughout the flow, the system handles background events (represented by the isolated "System" blocks in the visual diagram).

- **State Syncing:** When a doctor marks a patient as IN_CONSULTATION, the receptionist and queue displays must instantly update (via WebSockets/Server-Sent Events).
- **Automated Billing Triggers:** Prescribing a medicine or ordering a lab test must automatically append the cost to the patient's active draft invoice.
- **Audit Logging:** Every state change (e.g., Status changed to CHECKED_IN) must log the User_ID, Timestamp, and Action in the Audit DB.
- **Notification Triggers:** Booking success, report generation, and payment receipts trigger asynchronous workers (e.g., Celery/Redis queues) to send SMS/WhatsApp messages.

4. Developer Implementation Notes

- **State Management:** Utilize strict Enum types for the Visit State (SCHEDULED, CHECKED_IN, IN_CONSULTATION, WAITING_FOR_LABS, COMPLETED).
- **Data Passing:** Ensure the Visit_ID and UHID are passed contextually across all frontend routes to maintain the user session context without requiring re-fetching.
- **Error Handling:** Implement robust fallback UI states for when third-party systems (like payment gateways or SMS providers) fail during critical flow transitions.